

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

Develop Static Webpage Using HTML Components

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Course Outcome: CO1

1. TITLE

Design Static Webpage using HTML Components

2. AIM

To design and implement a static webpage using HTML components and semantic HTML5 elements to create a clear and organized document structure.

3. OBJECTIVE

- To understand the basic structure of an HTML5 document.
- To use semantic HTML5 elements such as header, nav, main, section, article, aside and footer.
- To create navigation links within a webpage.
- To organize webpage content using HTML tables and structural elements.
- To create an interactive form using HTML form controls.
- To use headings, paragraphs, lists, links and other HTML components.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to structure content on web pages. HTML elements define the meaning and organization of webpage content.

Semantic HTML5 elements provide meaningful structure to a webpage. Elements such as <header>, <nav>, <main>, <section>, <article>, <aside> and <footer> identify different parts of the document and make the structure easier for browsers, screen readers and other tools to understand.

HTML also provides components for navigation, text presentation, lists, tables and user input. Forms can contain controls such as text fields, email fields, selection lists, text areas and buttons.

5. PROCEDURE

1. Create an HTML5 document using the <!DOCTYPE html> declaration.
2. Define the document language and basic head information.
3. Create a header containing the webpage title and navigation links.
4. Create a hero section with a heading, description and article link.
5. Use main, article and section elements to organize the primary content.
6. Create a form using fieldset, legend, label, input, select and textarea elements.
7. Create an aside section containing references, benefits and required components.
8. Add a footer containing the webpage copyright information.
9. Save the file with an .html extension and open it in a web browser.
10. Verify that the webpage structure, links, form controls and content are displayed correctly.

6. HOW TO RUN THE PROGRAM

11. Open Visual Studio Code and create a new HTML file.
12. Paste the complete HTML source code into the file.
13. Save the file with an .html extension, for example index.html.
14. Open the HTML file in a web browser.
15. Verify the navigation links, articles, form controls, lists and footer.

7. PROGRAM

Complete HTML Source Code

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>Pure Semantic HTML Static Webpage</title>
</head>

<body>

  <!-- Header Section -->
  <header>
    <table width="100%" border="0" cellspacing="0" cellpadding="10">
      <tr>
        <td>
          <h1>StaticWeb</h1>
        </td>
        <td align="right">
          <nav>
            <a href="#home">Home</a> &nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&~
            <a href="#articles">Articles</a> &nbsp;&nbsp;&nbsp;&nbsp;&~
            <a href="#about">About</a> &nbsp;&nbsp;&nbsp;&nbsp;&~
            <a href="#contact">Contact Us</a>
          </nav>
        </td>
      </tr>
    </table>
  </header>

  <br>

  <!-- Hero / Showcase Section -->
  <section id="home">
    <center>
      <h2>Semantic Structures Without CSS</h2>
      <p>This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.</p>
      <p><a href="#articles"><b>Read Our Articles Below &darr;</b></a></p>
    </center>
  </section>

  <br>

  <!-- Main Content Layout using a Structural Table -->
  <table width="100%" border="0" cellspacing="0" cellpadding="20">
    <tr valign="top">

      <!-- Left Side: Main Content Column -->
      <td width="70%">
        <main id="articles">
```

```
<article>
  <h2>1. Core Structural Tags</h2>
  <p>Published: July 7, 2026</p>
  <p>When you omit cascading style sheets entirely, the semantic markup handles all of the
    structural heavy lifting. Elements like <code>&lt;main&gt;</code>,
    <code>&lt;article&gt;</code>, and <code>&lt;section&gt;</code> tell screen readers and web
    scrapers exactly how information flows.</p>
  <p>Using raw browser defaults ensures excellent loading performance, perfect accessibility
    compliance, and absolute structural clarity.</p>
</article>
<br>
<hr><br>
```

```
<article>
  <h2>2. Native Interaction Elements</h2>
  <p>Published: July 5, 2026</p>
  <p>Browsers come equipped with excellent interactive elements by default. We can capture text
    inputs, provide multiple-choice configurations, and submit structured information to backend
    parsers using pure markup.</p>
```

```
<h3>Interactive Demonstration Forms</h3>
<form id="contact" action="#" method="post">
  <fieldset>
    <legend><b>User Inquiry Form</b></legend>
    <p>
      <label for="name">Full Name: </label>
      <input type="text" id="name" name="username" required placeholder="John Doe">
    </p>
    <p>
      <label for="email">Email Address: </label>
      <input type="email" id="email" name="useremail" required>
    </p>
    <p>
      <label for="topic">Inquiry Topic: </label>
      <select id="topic" name="usertopic">
        <option value="general">General Feedback</option>
        <option value="technical">Technical Architecture</option>
        <option value="academics">Academic Curriculum</option>
      </select>
    </p>
    <p>
      <label for="message">Message Details:</label><br>
      <textarea id="message" name="usermessage" rows="5" cols="40"
        placeholder="Type your notes here..."></textarea>
    </p>
    <p>
      <input type="submit" value="Submit Form Data">
      <input type="reset" value="Clear Entries">
    </p>
  </fieldset>
</form>
</article>
```

```
</main>
</td>
```

```

<!-- Right Side: Sidebar Column -->
<td width="30%" bgcolor="#f4f4f4">
  <aside id="about">
    <h3>Document References</h3>
    <p>Static pages map data out directly without real-time database queries or processor overhead.</p>
    <h4>Core Benefits:</h4>
    <ul>
      <li>Instant browser painting</li>
      <li>Low memory footprints</li>
      <li>High data compatibility</li>
    </ul>
    <h4>Required Components:</h4>
    <ol>
      <li>Document Type Definition</li>
      <li>Semantic Wrapper Blocks</li>
      <li>Data Input Control Interfaces</li>
    </ol>
  </aside>
</td>

</tr>
</table>

<hr>

<!-- Footer Section -->
<footer>
  <center>
    <p>&copy; 2026 Pure HTML Architecture. Assembled completely without style configurations.</p>
  </center>
</footer>

</body>

</html>

```

8. CODE EXPLANATION

Code / Concept	Purpose
<!DOCTYPE html>	Declares the document as an HTML5 document.
<header>	Defines the introductory/header area of the webpage.
<nav>	Contains navigation links to sections of the page.
<section>	Groups related content into a logical section.
<main>	Identifies the main content of the webpage.
<article>	Represents independent article content.
<aside>	Contains related or supporting sidebar content.
<form>	Creates an area for collecting user input.
<fieldset> and <legend>	Groups form controls and provides a caption for the group.
<input>, <select>, <textarea>	Provide different types of user input controls.
<footer>	Defines the footer section of the webpage.

9. TEST CASES AND EXECUTION DATA

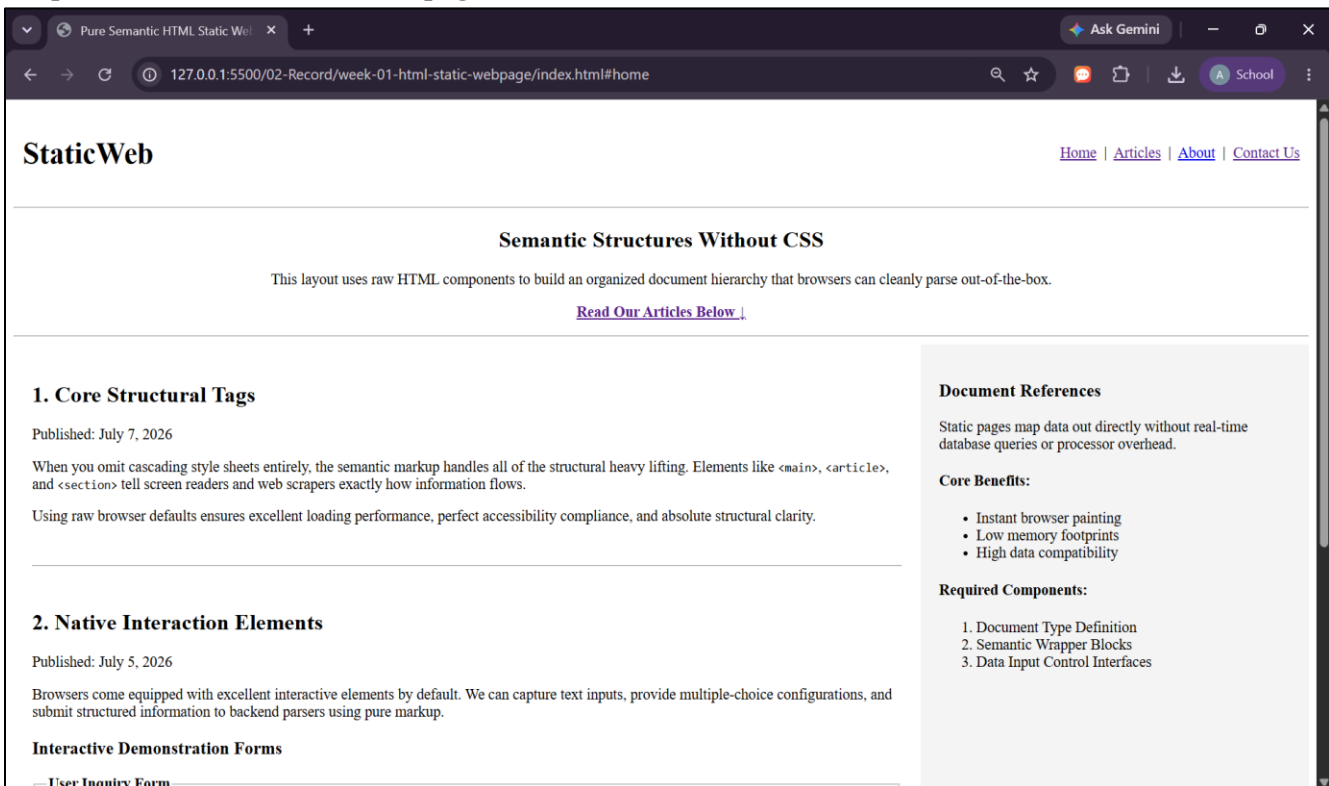
The following test cases can be used to verify the webpage after opening it in a browser.

Test Case	Test / Input	Expected Result	Passed
TC-01	Open HTML file	Webpage should load without structural errors.	Test case Passed
TC-02	Click Home / Articles / About / Contact Us	Navigation should move to the corresponding section.	Test case Passed
TC-03	Check semantic sections	Header, navigation, main, articles, aside and footer should be visible in the correct structure.	Test case Passed
TC-04	Enter Full Name and Email	The form should accept the entered values.	Test case Passed
TC-05	Select Inquiry Topic	The selected option should be displayed in the control.	Test case Passed
TC-06	Enter Message Details	Textarea should accept multiline input.	Test case Passed
TC-07	Click Clear Entries	Entered form values should be reset.	Test case Passed

10. OUTPUT

Paste the actual browser output screenshot below.

Output: Screenshot 1: Static Webpage



Screenshot 2: User Inquiry Form

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5500/02-Record/week-01-html-static-webpage/index.html#articles". The page content includes an introductory paragraph about semantic markup, a section titled "2. Native Interaction Elements" with a publication date of July 5, 2026, and a paragraph about browser defaults. Below this is a section titled "Interactive Demonstration Forms" containing a "User Inquiry Form". The form has fields for "Full Name" (filled with "Venu gopal"), "Email Address" (filled with "venu@gmail.com"), "Inquiry Topic" (a dropdown menu showing "Academic Curriculum"), and "Message Details" (a text area filled with "Teachers are teaching very well."). There are "Submit Form Data" and "Clear Entries" buttons at the bottom of the form. To the right of the main content is a sidebar with two sections: "Core Benefits:" listing "Instant browser painting", "Low memory footprints", and "High data compatibility"; and "Required Components:" listing "1. Document Type Definition", "2. Semantic Wrapper Blocks", and "3. Data Input Control Interfaces". The footer of the page reads "© 2026 Pure HTML Architecture. Assembled completely without style configurations."

When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like `<main>`, `<article>`, and `<section>` tell screen readers and web scrapers exactly how information flows.

Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.

2. Native Interaction Elements

Published: July 5, 2026

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

Message Details:

Core Benefits:

- Instant browser painting
- Low memory footprints
- High data compatibility

Required Components:

1. Document Type Definition
2. Semantic Wrapper Blocks
3. Data Input Control Interfaces

© 2026 Pure HTML Architecture. Assembled completely without style configurations.

11. RESULT

Thus, a static webpage was designed and implemented using HTML components and semantic HTML5 elements. The webpage includes navigation, structured articles, a sidebar, lists, an interactive form and a footer.

12. CONCLUSION

The experiment demonstrates the use of HTML5 structural and semantic elements to organize webpage content and create a functional static webpage without external styling.